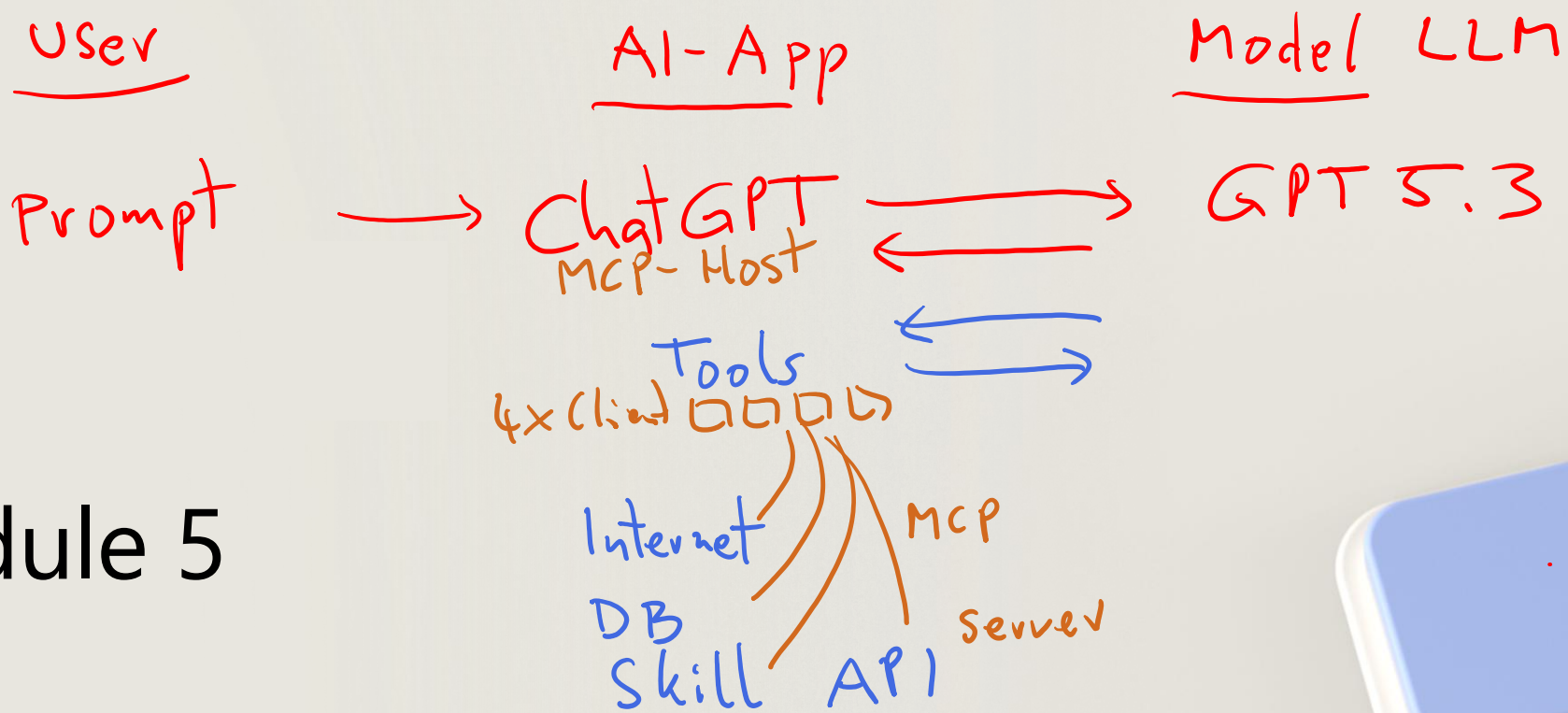




Module 5

Introduction to GitHub Copilot

MCP

Agenda



GitHub Foundations Part 1

- 1 Introduction to Git ✓
- 2 Introduction to GitHub ✓
- 3 Introduction to GitHub's products ✓
- 4 Configure code scanning on GitHub
- 5 Introduction to GitHub Copilot ←
- 6 Code with GitHub Codespaces ✓
- 7 Manage your work with GitHub Projects
- 8 Communicate effectively on GitHub using Markdown

GitHub Foundations Part 2

- 9 Contribute to an open-source project on GitHub
- 10 Manage an InnerSource program by using GitHub
- 11 Maintain a secure repository by using GitHub best practices
- 12 Introduction to GitHub administration
- 13 Authenticate and authorize user identities on GitHub
- 14 Manage repository changes by using pull requests on GitHub
- 15 Search and organize repository history by using GitHub
- 16 Using GitHub Copilot with Python

Introduction

GitHub Copilot draws context from comments and code to suggest individual lines and whole functions instantly.

By the end of this module, you'll:

- Understand how GitHub Copilot can help you code by offering autocomplete-style suggestions.
- Understand the various ways to trigger GitHub Copilot.
- Comprehend the differences among GitHub Copilot Free, Pro, Business, and Enterprise.
- Know how to configure GitHub Copilot.
- Know how to troubleshoot GitHub Copilot.

GitHub Copilot, your AI pair programmer



Copilot for Chat

- GitHub Copilot integrates a ChatGPT-like chat interface into VS Code and Visual Studio.
- It recognizes typed code and error messages, providing in-depth analysis and explanations.
- Developers can generate unit tests and receive proposed fixes for bugs.



Copilot for Pull Requests

- Utilizes OpenAI's GPT-4 model to add AI-powered tags in pull-request descriptions.
- Automatically fills out tags based on changed code, which developers can review or modify.
- Available through a GitHub app that can be installed by organization admins and repository owners.



Copilot for the CLI

- Assists developers in composing commands and loops in the terminal.
- Helps with remembering precise syntax and using obscure find flags.
- Enhances productivity by reducing the need to scroll through pages for command syntax.

Interact with Copilot



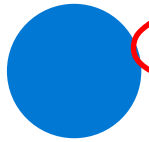
Inline suggestions

- To accept a suggestion, select the **Tab** key or the > (right arrow) key.
- To reject a suggestion, keep typing or select the **Esc** key.



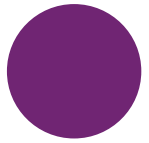
Command palette

1. Open the command palette by selecting **Ctrl+Shift+P** (Windows or Linux) or **Cmd+Shift+P** (Mac).
2. Enter Copilot to see available commands.
3. Select actions like **Explain This** or **Generate Unit Tests** to get assistance.



Copilot chat

1. Open the Copilot chat panel in your IDE.
2. Enter questions or requests in natural language, and then evaluate the Copilot response.



Comments to code

When you select the **Enter** key, Copilot generates code based on your description.

```
// Function to reverse a string
def reverse_string(s):
    return s[::-1]
```

Set up, configure, and troubleshoot GitHub Copilot

Signing Up for GitHub Copilot

- Set up a free trial or subscription for your account via GitHub settings under "Code, planning, and automation."
- Install an extension for your preferred environment (supports GitHub.com, VS Code, Visual Studio, JetBrains IDEs, and Neovim).

Configuring GitHub Copilot in VS Code

- Install the GitHub Copilot extension from the Visual Studio Marketplace and open VS Code.
- Sign in to GitHub in VS Code if not previously authorized.
- Enable or disable GitHub Copilot and configure advanced settings within VS Code for code autocompletion.

Exercise - Develop with AI-powered code suggestions by using GitHub Copilot and VS Code



GitHub Copilot helps by offering autocomplete-style suggestions.

Learn how GitHub Copilot works, and what to consider while using GitHub Copilot.

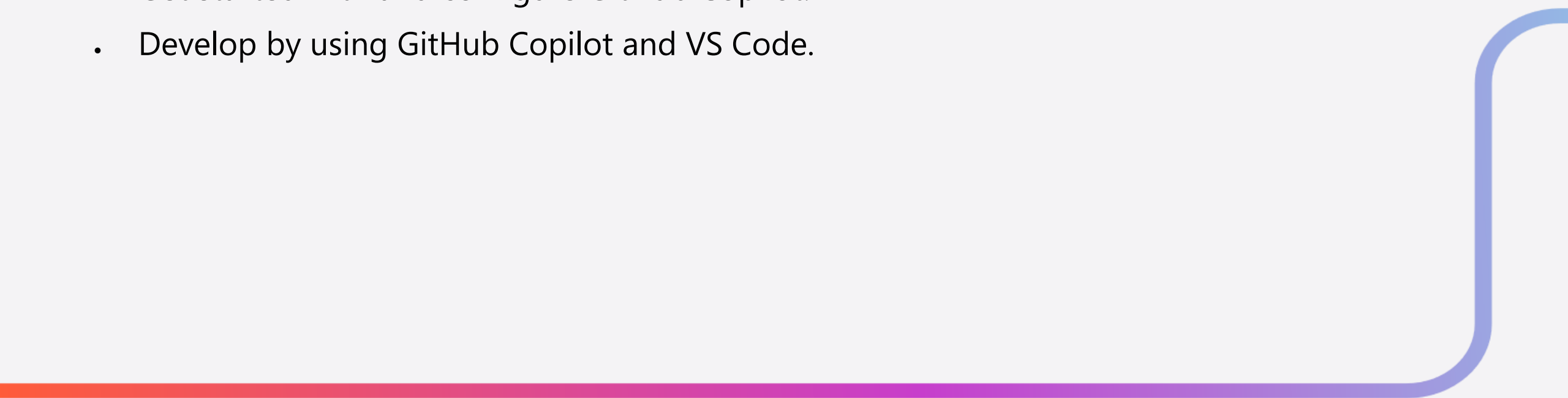
This exercise walks you through the following steps:

- Install GitHub Copilot by using GitHub Codespaces
- Prompt GitHub Copilot for code suggestions
- Accept code suggestions from GitHub Copilot

 [Start course](#)

Summary

Now that you've finished this module, you should be able to:

- Explain the GitHub Copilot features and the subscription plans (Free, Pro, Business, and Enterprise).
 - Get started with and configure GitHub Copilot.
 - Develop by using GitHub Copilot and VS Code.
- 

End of presentation



